Iteration C4: Functional Testing of Controllers

Now for the moment of truth. Before we focus on writing new tests, we need to determine if we've broken anything. Remembering our experience after we added validation logic to our model, with some trepidation we run our tests again:

```
depot> bin/rails test
```

This time, all is well. We added a lot, but we didn't break anything. That's a relief, but our work isn't done yet; we still need tests for what we just added.

The unit testing of models that we did previously seemed straightforward enough. We called a method and compared what it returned against what we expected it to return. But now we're dealing with a server that processes requests and a user viewing responses in a browser. What we need is *functional* tests that verify that the model, view, and controller work well together. Never fear—Rails has you covered.

First, let's take a look at what Rails generated for us:

rails7/depot_d/test/controllers/store_controller_test.rb

```
require "test_helper"

class StoreControllerTest < ActionDispatch::IntegrationTest
  test "should get index" do
    get store_index_url
    assert_response :success
  end
end
```

The `should get index` test gets the index and asserts that a successful response is expected. That certainly seems straightforward enough. That's a reasonable beginning, but we also want to verify that the response contains our layout, our product information, and our number formatting. Let's see what that looks like in code: